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# 4 ACCELERATION DUE TO GRAVITY USING SIMPLE PENDULUM

**AIM**

To measure the acceleration due to gravity using a simple pendulum

**APPARATUS REQUIRED**

Retort stand, pendulum bob, thread, meter scale, stop watch.

**FORMULA**

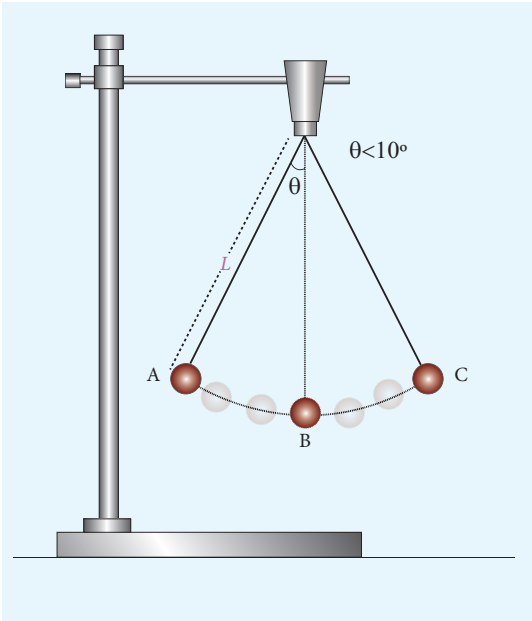
Acceleration due to gravity $g = 4\pi^2 \left( \frac{L}{T^2} \right)$ (m $s^{-2}$)

where T → Time period of simple pendulum (second)

g → Acceleration due to gravity (metre $sec^{-2}$)

L → Length of the pendulum (metre)

**DIAGRAM**




**PROCEDURE**

- Attach a small brass bob to the thread
- Fix this thread on to the stand
- Measure the length of the pendulum from top to the middle of the bob of the pendulum. Record the length of the pendulum in the table below.
- Note the time (t) for 10 oscillations using stop watch
- The period of oscillation $T = \frac{t}{10}$
- Repeat the experiment for different lengths of the pendulum 'L'. Find acceleration due to gravity g using the given formula.